

Steady State Library for Liquid Rocket Engine Cycle Design

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The use of system modelling tools for design and analysis of liquid rocket engines is necessary during the pre-design activities; their flexibility makes them a fundamental instrument along the entire design period from pre-design phase to parametric studies for fine tuning of the engine parameters. For this purpose these tools have to be reliable and fast, in order to perform iterative engine design loops and parametric studies in a reasonable time. Moreover, they should be able to perform off-design predictions and detailed analyses.

The Steady State library presented within this paper enables to realize these purposes. Its object oriented structure ensures maximum flexibility and allows for studying any liquid rocket engine thermodynamic cycle (open or closed).¹² In this paper the tool is validated with the RL-10 engine⁴ and its capability will be demonstrated by designing several engine cycles.

The entire library has been implemented within the existing analysis software Ecosim-Pro, designed to model various kinds of dynamic systems. The standard propulsion library ESPSS⁶ can be used to study both stationary states and transients of a propulsion system.⁸ Unfortunately its use for steady state applications is not trivial because of the complexity of the transient models there implemented. Therefore, simplified pre-design and parametric studies are difficult and time-consuming.

The library described in this paper has been designed specifically for steady state purposes and can be considered as a completion of the ESPSS library, providing a helpful and fast tool for the pre-design phase and off-design analyses.

To this aim, the available fluid properties and combustion modelling functions of ESPSS have been implemented in an adequate form into the new libraries. Additionally, fluid dynamic, combustion and heat transfer models have been developed to simulate the physical steady state behaviour of the main components of a propulsion system, as pipes, valves, turbines, pumps, orifices, combustion chamber and nozzle. These components are suited for both launcher and spacecraft applications.

The main components feature a flag that enables the user to switch from “design” mode to the “off-design” analysis mode to enhance the flexibility of the library and perform different simulations with the same model.

After the description of the main components of the Steady State library, their integration and validation will be presented through different examples of liquid rocket engine cycles design.

I. Introduction

The European Space Propulsion System Simulation libraries (ESPSS) have been developed by Iberespacio in the frame of two ESA contracts in the last 5 years, and enable the modelling and analysis of propulsion systems for both spacecraft and launcher applications.

A new library has been developed to enable EcosimPro steady state design and off-design analysis of liquid propulsion systems. The final aim of this study is to obtain a library able to perform design and parametric analysis of liquid propulsion systems, whose components can act as seamless replacements for the transient ESPSS ones.

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II. Models

The Steady State library is based on the following libraries of EcosimPro and ESPSS:

- **EcosimPro:**

- MATH library
- CONTROL library
- PORTS_LIB library
- THERMAL library

- **ESPSS:**

- FLUID_PROPERTIES
- FLUID_FLOW_1D (partially, only for friction factor correlations)
- COMB_CHAMBERS (partially, for combustion products functions)

In particular, the fluid properties functions and the chemical equilibrium functions for combustion are taken from ESPSS, as they serve the same purpose needed for the steady state. The fact that they are more complex (enabling the use of two fluids at the same time, for instance, which is not needed in the Steady State library) just adds some complexity to the readability of the code, but brings no compatibility issues nor noticeable slowdowns of the simulations. Figure 1 gives an overview of the components developed specifically for the Steady State library. The main components are described hereafter.

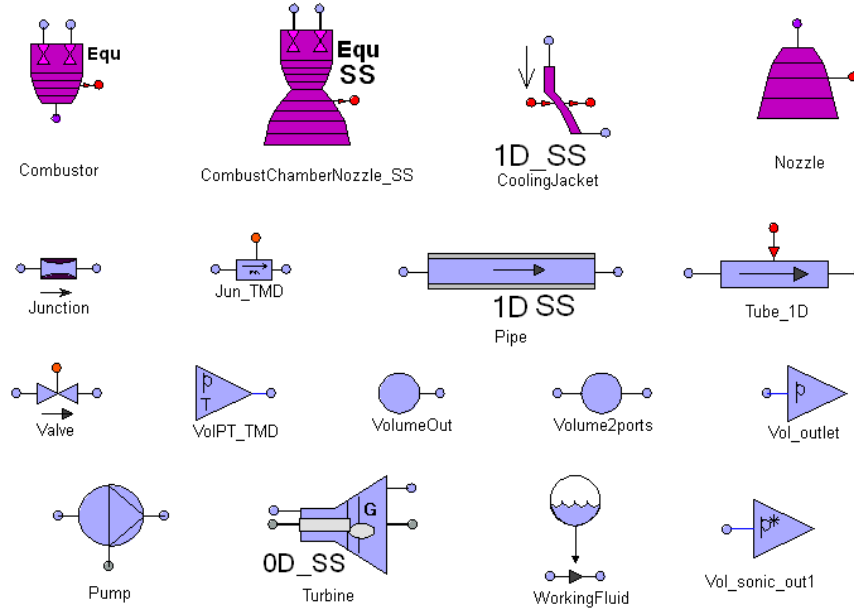


Figure 1. Steady state library: components overview

A. Ports

Ports are used to connect a component to another one or to different ones, in order to guarantee the propagation of the variables. Two new ports have been created: the Steady State *fluid* port,⁷ that represents

the basis and the rationale on which the whole library is coded, and the *nozzle* port, to better manage nozzle connections, similarly to the transient nozzle port.

Each port can connect two or more components at once. SUM variables, such as mass flow rate, will be summed at the ports to ensure mass flow conservation; EQUAL variables, such as stagnation pressure, will be propagated to all components connected at the same port. A standard flow component should have two ports, one IN and one OUT, thus defining the mass flow direction. IN ports ensure calculation of the enthalpy flow mh , while enthalpy h is computed in the OUT ports.

Similarly to the original ESPSS fluid port, the Steady State fluid port can propagate the molar fraction of chemical species to allow the correct evaluation of fluid flow of combustion products in systems where this is required (e.g. staged combustion cycles).

B. Functions

Several functions present in ESPSS are used in the Steady State library. Most of them are used to calculate friction factors inside the pipes, combustion products inside combustion chambers or to define the geometrical mesh of a 1-D component. The authors would like to point to Ref.⁶ and Ref.⁹ for the details of the ESPSS implementation. Some functions had to be modified or rewritten.

C. The type switch

Most components have a type switch, that enable switching the model between Design and Off-Design mode.

- **Design mode.** Geometrical construction data for junctions and valves are an output of the components. They are calculated from a given ΔP .

The combustion chamber requests chamber pressure, mixture ratio MR and throat diameter as main design inputs. The propellant mass flows and heat fluxes are its main outputs.

Turbomachinery components in design mode evaluate performance parameters as torque, power and rotational speed using the mass flow and pressures coming from the ports.

- **Off-Design mode.** This mode can be used for the analysis of a given cycle with fixed geometry and main characteristics. Here, junctions and valves have a given geometry. Mass flow or ΔP are calculated, depending on the relative placement of the components.

Combustion chambers have a given nozzle throat diameter (as in design mode), but mass flows are given from the inlet ports. Chamber pressure and MR are calculated accordingly.

As the reader can imagine from this general description, some additional options are given to the user in Design mode, in order to fine-tune the models depending on the cycle studied. Therefore, mass flow ratios through splits can be user-fixed (inputs), or calculated as outputs. Likewise, turbine pressure ratios can be fixed or calculated from the given cycle.

D. Tube, pipe and cooling jacket components

The tube component is able to evaluate a one-dimensional flow in steady state conditions. The tube takes into account the enthalpy variation due to external heat fluxes and the pressure drop due to the friction along the pipe. As in the transient version of the component, the Steady State tube is divided in volumes and junctions. Pressure drops and enthalpy variations are calculated at the end of each volume, on the junction. The governing equations are the following:

- **Mass conservation**

$$\dot{m}_{in} = \dot{m}_{out} = \rho v A \quad (1)$$

- **Momentum conservation**

$$P_{i+1} = P_i - \Delta P_i \quad (2)$$

where, for each node i ,

$$\Delta P = \frac{1}{2} \zeta \frac{\dot{m}^2}{\rho A^2} \quad (3)$$

$$\zeta = \frac{\Delta L_i}{D_h} fr \quad (4)$$

and where the friction factor fr is a function of Re and relative roughness, as defined in Ref.⁹ In order to ensure numerical stability, fr is limited in the range $[10^{-2}, 1]$, enabling flows with Reynolds numbers between 64 and 10^8 .

- **Energy conservation.** For each node i :

$$\dot{m} \Delta H_i = \dot{q}_w \Delta A_{wet,i} \quad (5)$$

The heat flux is evaluated using the thermal port and connecting the component with all components inside the EcosimPro THERMAL library. It has been decided to use the original transient EcosimPro THERMAL library, but still allowing to be interfaced with the Steady State library. This choice enables a relaxation in the overall steady state model stiffness thanks to the first order differential equations present in the THERMAL components. Of course, due to the specifications of the Steady State library, the time variable will have no physical meaning anymore, and it must rather be regarded as an integration constant.

The pipe component is inherited from the tube. Additionally it features a 1D wall model for the evaluation of the heat fluxes and heat capacities. As in the corresponding transient component, it includes a material pipe thermally connected to the tube, and permits simple convection with the ambient by using a constant convective coefficient h_c .

Cooling jacket component features two different models depending on the complexity of the propulsion system:

- cooling jacket
- regenerative circuit

The “cooling jacket” component is inherited from the tube component. It permits the modelling of a combustion chamber cooling jacket. Mass flow, pressure drop and temperature rise of the coolant are evaluated using the same governing equations as for the pipes and the tubes. Since the direction of the flow in the Steady State library must be given at the schematic design stage, two components are foreseen: a co-flow and a counterflow cooling jacket. They are constructed by aggregation of one tube representing the channels and a simplified 3D geometry (built by means of several bars around the channels) around them (see Figure 2).

The “regenerative circuit” component features a pre-defined pressure drop and a pre-assigned hot gas side wall temperature $T_{w,hot}$ profile along the combustion chamber. Mass flow value is coming from the ports; the component sends the $T_{w,hot}$ variable to the combustion chamber through the thermal port. In this way it is possible to evaluate the chamber heat flux $\dot{q}_{wall}$ (see Eq. 6). Using this variable, the enthalpy rise along the channel is evaluated and so the outlet temperature. Choosing a material for the chamber wall, the chamber wall thickness is an output of the design (see Eq. 7). The channel height is a user given input for the model, while the channel width is evaluated from the fixed number of channels.

$$\dot{q}_{wall} = h_c A_{wet} (T_{aw} - T_{w,hg}) \quad (6)$$

$$\dot{q}_{wall} = \frac{\lambda}{t} A_{wet} (T_{w,hg} - T_{w,cf}) \quad (7)$$

E. Thrust Chamber

The thrust chamber is composed by two different components following the same idea developed in the transient library. A combustor component and a nozzle component are linked together to create the thrust chamber.

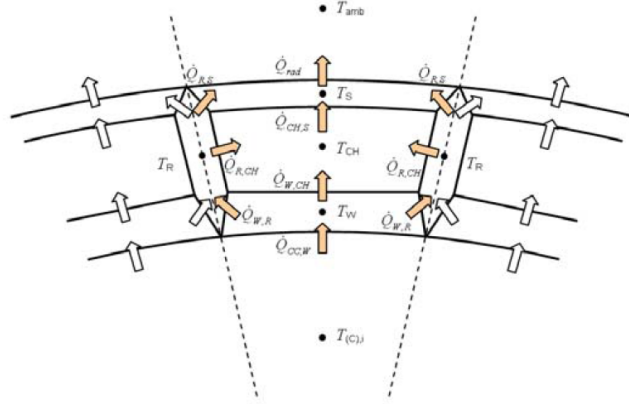


Figure 2. Cooling jacket channels wall mesh⁹

1. Combustor

This component represents a non adiabatic 1-D combustion process inside a convergent chamber (up to the throat section). It is a steady state, one dimensional, isenthalpic combustion chamber. The equilibrium combustion products are calculated using the minimum Gibbs energy method¹⁰ as a function of the propellant's mixture molar fractions and enthalpies and of the chamber pressure.

The chamber geometry allows for precise chamber contour definitions and non homogeneous node discretisation.

Thermodynamic properties along the chamber sections are evaluated using isenthalpic correlations in frozen conditions. Heat fluxes are calculated with the Bartz correlation in closed form.³ The chamber works only in “ignited” mode, as it is not required for a steady state model to show transitions between non burning and burning state.

The compositions of both fuel and oxidizer are evaluated from the injected fluids. It is possible to use either pure fluids or combustion products from a previous combustor (preburner). Along with the Design/Off-Design type switch, another switch is responsible for choosing the combustor type, which can be either a Main Combustion Chamber (MCC) or a Preburner (PB_GG). The main difference between the two combustor types is the following:

- MCC. For a Main Combustion Chamber in Design mode, chamber pressure is user given ($P_c = P_{design}$)
- PB_GG. For a Preburner in Design mode, chamber pressure is taken from the outlet port ($P_c = f_{out}.P$)

In Design mode, the combustor component calculates inlet mass flows from given chamber pressure, MR and nozzle throat diameter. The mass flow conservation is written as:

$$\dot{m} = \frac{\rho_{th} v_{th} A_{th}}{\eta_c} \quad (8)$$

where the subscript *th* refers to throat conditions, and η_c is the combustion efficiency.

In Off-Design mode, the component evaluates the equilibrium composition in the first section to obtain thermodynamic and transport properties, and in the throat to evaluate the chamber pressure and the mass flow rate by an iterative loop. This equation is actually used in the overall loop to determine the chamber pressure P_c implicitly. The ideal gas equation is written twice, for stagnation chamber and for throat conditions. Isentropic throat conditions are calculated iteratively assuming shifting equilibrium and variable isentropic coefficient γ .

For each node i , the relevant characteristics (Mach number M_i , P_i , T_i , ρ_i , sound speed $v_{sound,i}$, v_i) are calculated assuming isentropic flow conditions. This simplification is acceptable since these variables are only needed for assessing the heat transfer coefficient within the Bartz equation.

2. Nozzle

The component represents a 1D supersonic nozzle in steady state conditions. The choked throat conditions (P_{th} , T_{th} and v_{th}) are evaluated in the combustor component and communicated through the nozzle port. Stagnation conditions are calculated from the throat conditions. Static conditions are evaluated in each section using isentropic correlations and assuming frozen chemistry. The heat flux in each section is evaluated using the semi-empirical correlation of Bartz in a closed form.

F. Pump

The pump component features a simple model using isentropic relations and constant, user-given efficiency η_p to calculate pump conditions. The isentropic enthalpy rise is calculated assuming an isentropic transformation between inlet and outlet pressure. The Design type parameter decides whether the pump pressure rise is fixed from the ports (in Design mode) or calculated (in Off-Design mode). In both modes, given a assigned specific speed N_s , the shaft rotation speed ω is calculated. In Design mode, the inlet entropy s is calculated from inlet pressure P_{in} and enthalpy h_{in} . Then, the isentropic enthalpy rise dh_{is} is calculated with an isentropic transformation, with known inlet and outlet pressures P_{in} and P_{out} , and inlet entropy s_{in} . Finally, the real enthalpy rise dh is calculated as:

$$dh = \frac{dh_{is}}{\eta_p} \quad (9)$$

and subsequently, the shaft rotational speed ω and torque τ are then linked with the actual enthalpy rise by the power balance equation:

$$\omega \tau = \dot{m} \Delta h \quad (10)$$

In Off-Design mode, the inlet entropy s_{in} is calculated (as for the Design mode) from inlet pressure P_{in} and enthalpy h_{in} . The real enthalpy rise dh is given by the power balance equation, and the isentropic enthalpy rise dh_{is} is given by:

$$dh_{is} = \eta \cdot dh \quad (11)$$

Therefore, the outlet pressure P_{out} is calculated with an isentropic transformation, with known isentropic enthalpy rise dh_{is} and entropy s . As for the Design case, it is possible to compute torque and power from it (thank to an assigned specific speed N_s).

In the near future the off-design mode (with assigned specific speed) will be upgraded with the capability of using performance maps for efficiency and pressure head.

G. Turbine

The turbine component is a simple component using isentropic relations and constant, user-given efficiency η_t to calculate turbine conditions. The isentropic enthalpy fall is calculated assuming an isentropic transformation between inlet and outlet pressure. The construction parameter *turbine_type* decides, in Design mode, whether the mass flow $\dot{m}$ is assigned from the ports (known_mflow) thus calculating the upstream pressure, or the pressures are assigned from the ports (known_pressures, known_PI_tt), thus calculating the mass flows. This switch must be carefully set depending on the cycle studied:

- **turbine_type = known_mflow.** For closed cycles, in most cases, the mass flow is determined by the preburner (staged combustion) or by bypass valves (expander), therefore it is given from the ports, and the pressure ratio should be calculated in the turbine component.
- **turbine_type = known_pressures.** For open cycles (gas generator) or for one of the turbines in closed cycles with parallel turbines, the pressures should be fixed (known_pressures), and the model will find the mass flow that equilibrates the pump power.
- **turbine_type = known_PI_tt.** For open cycles, especially in the pre-design phase to have a first attempt result and to facilitate parametric studies changing the pressure ratio Π_{tt} ; the model will find the mass flow that equilibrates the pump power.

In all working modes (Design or Off-Design, known_mflow or known_pressures, known_Pi_tt), torque, shaft rotation speed ω and power are given from the mechanical port.

- **Design mode**

- **known_mflow.** The mass flow is used to calculate the real enthalpy fall (Eq. 12).

$$dh = -Power/\dot{m} \quad (12)$$

From this value, the isentropic enthalpy fall is calculated with the fixed efficiency (Eq. 13).

$$dh_{is} = dh/\eta \quad (13)$$

The inlet entropy s_{in} is calculated from the outlet pressure P_{out} and the ideal outlet enthalpy $h_{out} = h_{in} - dh_{is}$. Then, the inlet pressure is calculated from entropy s and inlet enthalpy h_{in} .

- **known_pressures.** The inlet entropy s_{in} is calculated from the inlet pressure P_{in} and the inlet enthalpy h_{in} . Then, the isentropic enthalpy fall dh_{is} is calculated knowing the inlet enthalpy h_{in} , the outlet pressure P_{out} and the inlet entropy s_{in} (Eq. 14).

$$dh_{is} = h_{in} - f(s_{in}, P_{out}) \quad (14)$$

Thereafter, the real enthalpy fall is given by using the turbine efficiency (Eq. 15).

$$dh = \eta \cdot dh_{is} \quad (15)$$

The mass flow is finally estimated from Eq. 12.

- **known_PI_tt.** When this turbine type is chosen, the calculation procedure differs from the one used in known_pressures only because the inlet pressure P_{in} is given as an output from P_{out} times Π_{tt} .
- **Off-Design mode.** The procedure is roughly the same as the Design mode case with turbine_type = known_pressures.

III. Design Procedure

At the beginning of a design analysis, a set of performance parameters must be chosen as assumption to define the engine class and the initial condition of the engine:

- Propellants
- Tank pressure and temperatures
- Chamber Pressure P_c
- Chamber Mixture Ratio MR
- Combustion efficiency η_{c*}
- Throat Diameter D_t
- Pump efficiencies η_p
- Pump specific speeds N_s
- Turbine efficiencies η_t

Subsequently, it is possible to evaluate other important characteristics such as the contour of the chamber (by use of L^* and simple geometrical correlations) and injector pressure drops. Using these variables it is now possible to build an engine schematic in the EcosimPro platform.

The schematics shown hereafter are only the graphical interfaces of mathematical models where all variables of each component are considered. The tool is able of calculating the steady state of the mathematical model by solving the non-linear algebraic equation system that results from the built schematic by means of the “Newton-Raphson” or the “Minpack” method.

In order to have a flexible and robust tool, each component model is carefully coded to provide the most suitable variables that can be used to break the non-linear algebraic loops. The choice of the correct variables that enable the equations system to converge represents one of the most important achievements of this work.

IV. Results

Several test cases have been performed in order to evaluate the reliability of the Steady State library. Following a step by step approach, it was decided to first validate each component singularly and then more complex systems. The results of this work have been presented last year at the Space Propulsion Conference 2010.⁷

The work showed a preliminary version of the Steady State library and model validations at component and subsystem level: pipelines, combustion chamber and turbopumps components have been validated, as well as turbopump and combustion chamber subsystems of the HM7B engine, the upper stage engine of the Ariane 5 ECA launcher. These two subsystems have been used to validate the models in Off-Design mode, comparing the steady state simulation results with the ones obtained during the ESPSS Industrial Evaluation⁸ from Astrium Bremen to validate the ESPSS library.

The purpose now is to validate the Steady State library for designing complete liquid rocket engine cycles.

A. HM7B rocket engine system

The HM7B is a gas generator cycle with single turbine and geared pumps. The two subsystems modelled in the previous work have been updated to the current library implementation and linked together to build the HM7B engine system model.

Figure 3 shows the schematic of the HM7B engine using the Steady State library. All model components are in *Design mode*.

In Table 1 the main input data for the system is collected; In Table 2 the main system variables results are summarized and compared with nominal data. Where nominal values are shown, they are taken from the open literature.

Pressure drop has been fixed in each valve and junction as well as turbomachinery efficiency and gas generator mixture ratio. For the turbine, the “known_PL_{tt}” type is chosen, while the pump specific speed N_s has been calculated from design data.

The main chamber pressure and mixture ratio are fixed. Propellant mass flows to the main chamber are calculated, and fed back to the upstream components. The gas generator mass flow is not fixed. Only its mixture ratio is fixed, in accordance with the maximum allowable temperature in the turbine. The mass flow is then calculated by an algebraic equation system resulting automatically from the connection of gas generator and turbopumps. The needed shaft power drives the total gas generator mass flow rate, since the turbine pressure ratio is fixed by design.

Initial values such as mass flow rates and shaft speed are chosen by rough engineering assessments. The robustness of the Steady State library is demonstrated by the stability of the simulation in a wide range of initial conditions.

From Table 2 a very good agreement with nominal data is recognisable. Few parameters have an higher percentage error: the fuel mass flow rate in the pump does not take into account the dump and the tap-off mass flow rate vented from the engine. The cooling jacket exit temperature takes into account the temperature increase in the injector dome. If compared to the LH2 injector dome temperature, the error decreases to 2.47%. Turbomachinery parameters show quite good results; the differences are mainly due to the turbine inlet temperature that is lower then the nominal one.

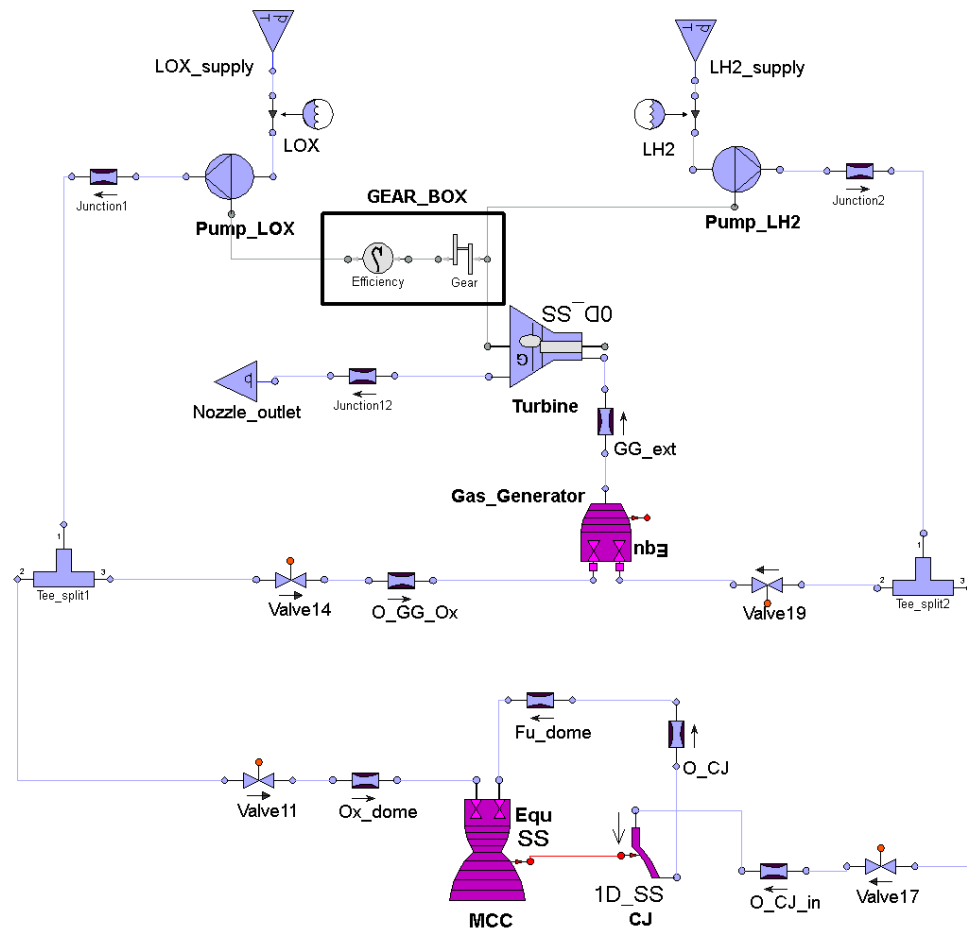


Figure 3. HM7B engine system schematic

Table 1. HM7B input¹ and initial data

Name	Description	Value	Units
$P_{in,LOX}$	Total pressure in LOX tank	2.0	bar
$T_{in,LOX}$	Total temperature in LOX tank	91.2	K
$P_{in,LH2}$	Total pressure in LH ₂ tank	3.0	bar
$T_{in,LH2}$	Total temperature in LH ₂ tank	21.0	K
P_{cc}	Nominal chamber pressure	36.6	bar
MR	Nominal chamber mixture ratio	5.	-
n_{ch}	Numbers of channels	128	-
$T_{i,cc}$	Initial chamber temperature	1000	K
P_o	Initial total pressure in the channels	50	bar
T_o	Initial total temperature in the channels	30	K
m_o	Initial mass flow in the channels	2	kg/s
$P_{c,gg}$	Initial Gas Generator pressure	20	bar
T_{ch}	Initial Gas Generator temperature	900	K
rpm_{ox}	Initial LOX pump speed	1000	rpm
$N_{s,ox}$	LOX pump specific speed	10.95	-
rpm_{fu}	Initial H ₂ pump speed	6000	rpm
$N_{s,fu}$	H ₂ pump specific speed	9.29	-
$m_{o,tu}$	Initial Turbine mass flow	0.2	kg/s

B. RL-10A-3-3A rocket engine system

The RL-10 is an expander cycle with single turbine and geared pumps. Being a closed cycle, the cycle design is more difficult than in an open cycle, because all parameters are strongly dependent from each other.

In this test case a model of the RL-10A-3-3A rocket engine system has been created and simulated in *Design* mode. Model results from the steady state calculations have been compared with the typical engine performance parameters at nominal operating conditions.^{2,5}

Pressure drops of the main valves and junctions are taken from engine typical values⁵ as well as for the tank pressure and temperature conditions. Pumps and turbine efficiency are constant and taken from open literature.^{2,5}

No calibration has been adopted for the control valves: the Oxidiser Control Valve (OCV) aperture ratio has not been trimmed because the mixture ratio is assigned in the combustion chamber. Mass flows are given by combustion chamber conditions. Turbomachinery power is regulated by the Thrust Control Valve (TCV) that is open at its nominal open area ratio of 9%.¹¹

In Table 3 the main input data for the system is collected;

In Table 4 the main system variables results are summarized, performed by the steady state model and compared with performance parameters at nominal condition.

The chamber pressure is assigned together with the mixture ratio. The pressure cascade for both propellant lines is calculated by the model according to design parameters such as valve pressure drops. The pressure rise for the LOX pump results directly from the calculated LOX pressure cascade, whereas an algebraic loop is solved for calculating the needed H₂ pump pressure rise and the turbine pressure ratio. In parallel, several other algebraic loops are solved, determining key variables such as cooling channel outlet temperature and turbine mass flow.

The nominal fuel bypass flow ratio¹¹ has been fixed using the split component. The turbine evaluates the required pressure ratio and the mass flow is then evaluated via the Thrust Control Valve (TCV).

The cooling channel component evaluates iteratively with the combustion chamber component the heat fluxes and the wall temperatures. Finally, pumps power is evaluated from the required pressure rise and mass flow, and rotation speed from the given specific speed N_s .

The RL10 design model here described has proven to be very robust with respect to initial conditions. The

Table 2. HM7B engine system output data

Name	Nominal Value ¹	Error
$\dot{m}_{ox,gg}$ [kg/s]		0.59%
$\dot{m}_{fu,gg}$ [kg/s]		0.23%
$\dot{m}_{t,gg}$ [kg/s]		0.3%
P_{gg} [bar]		0%
T_{gg} [K]		1.03%
$m_{cc,ox}$ [kg/s]		0.88%
$m_{cc,fu}$ [kg/s]		0.83%
$m_{cc,t}$ [kg/s]	14.86	0.87%
m_{ch} [kg/s]		2.36%
ΔP_{ch} [bar]		0.5%
$T_{out,ch}$ [K]		6.7%
ω_t [rpm]	60500	0.16%
τ_t [N·m]		4.48%
W_t [W]		4.32%
$T_{in,t}$ [K]		11.3%
$\dot{m}_{p,LOX}$ [kg/s]		0.96%
ΔP_{LOX} [bar]	48	0.41%
$\omega_{p,ox}$ [rpm]	13000	0.16%
$\tau_{p,ox}$ [N·m]		1.39%
$\dot{m}_{p,LH2}$ [kg/s]		7.56%
ΔP_{LH2} [bar]	52	1.29%
$\omega_{p,fu}$ [rpm]	60500	0.16%
$\tau_{p,fu}$ [N·m]		6.0%

Table 3. RL-10A-3-3A input and initial data

Name	Description	Value	Units
$P_{in,LOX}$	Total pressure in LOX tank	2.43	bar
$T_{in,LOX}$	Total temperature in LOX tank	97.056	K
$P_{in,LH2}$	Total pressure in LH ₂ tank	1.86	bar
$T_{in,LH2}$	Total temperature in LH ₂ tank	21.44	K
P_{cc}	Nominal chamber pressure	32.75	bar
MR	Nominal chamber mixture ratio	5.055	-
n_{ch}	Numbers of cooling channels	180	-
$T_{i,cc}$	Initial chamber temperature	1000	K
m_o	Initial mass flow in the channels	2	kg/s

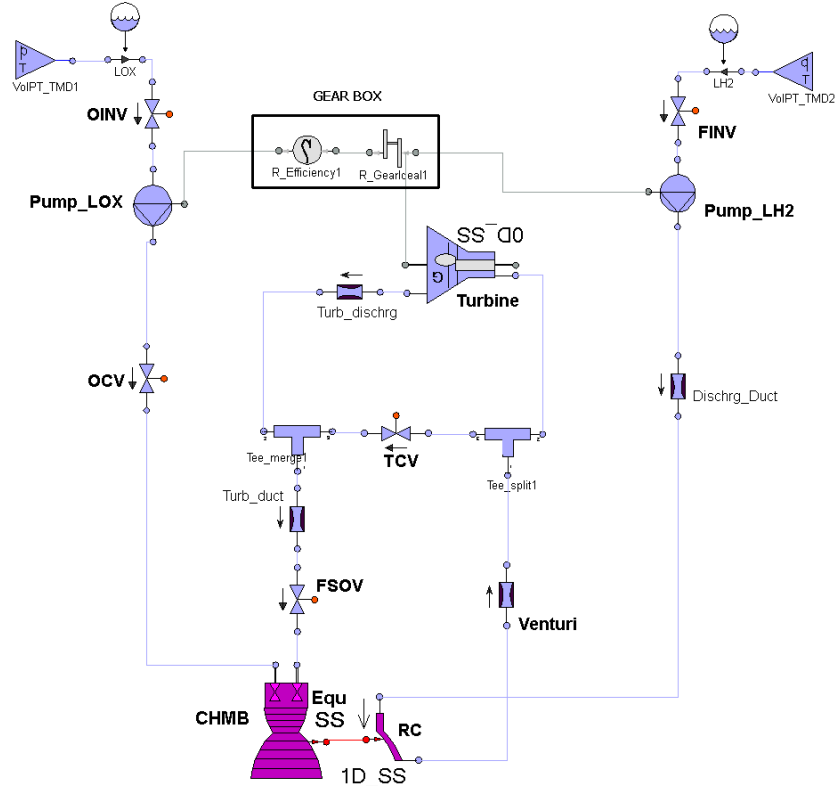


Figure 4. Schematic of the RL-10 engine

Table 4. RL-10A-3-3A engine system output data

Name	Nominal Value	Error
$m_{cc,ox}$ [kg/s]	13.95	0.38%
$m_{cc,fu}$ [kg/s]	2.76	0.38%
$m_{cc,t}$ [kg/s]	16.71	0.38%
m_{ch} [kg/s]	2.76	0.38%
$T_{out,ch}$ [K]	213.44	0.18%
ω_t [rpm]	31 537	0.39%
$\dot{m}_t$ [kg/s]	2.69	1.1%
W_t [kW]	588.36	3.85%
$\dot{m}_{p,LOX}$ [kg/s]	13.95	0.38%
$\omega_{p,ox}$ [rpm]	12 948	0.4%
$W_{p,ox}$ [kW]	82.026	4.07%
$\dot{m}_{p,LH2}$ [kg/s]	2.7946	0.85%
$\omega_{p,LH2}$ [rpm]	31 537	0.39%
$W_{p,LH2}$ [kW]	501.86	3.68%
$Thrust$ [kN]	73.4	2.89%
Isp [s]	444	2.5%

comparison of initial conditions in Table 3 and results in Table 4 demonstrates this assertion. For example, the initial mass flow in the cooling jacket m_o is 2 kg/s, whereas the simulation result yields 2.76 kg/s.

From Table 4 a very good agreement with nominal values is recognisable.

V. Conclusions

A new and improved EcosimPro library for steady state applications has been presented and validated in this paper. This library enables to perform in a fast and reliable way design and parametric analyses of liquid propulsion systems.

The present work describes a complete set of components able to perform dimensioning design studies and off-design analyses of liquid propulsion systems. A gas generator and an expander cycle have been chosen to validate the design capability of the steady state library. The resulting designs have been compared with actual liquid rocket engine data.

The steady state results when compared with nominal values show a good agreement as a proof of the accuracy of the library.

Future improvements of the Steady State library are foreseen. First of all, a more detailed model for the regenerative circuit component in design mode is under development, allowing the user to fix a reduced set of parameters. A second topic of improvement will be the refinement of the off-design mode for each model, enabling for instance the use of performance maps for turbomachinery components.

Acknowledgments

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